

Tutorial Sheet 1 (Limits, Continuity, Chain Rule, Change of Variables, and Jacobian)

1. Using the $\delta - \varepsilon$ approach, show that

a. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} = 0$

c. $\lim_{(x,y) \rightarrow (1,1)} (x^2 + y^2 - 1) = 1$

b. $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) = 0$

2. Evaluate the following limits (if exist)

a. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2}$

e. $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) \sin\left(\frac{1}{xy}\right)$

b. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2}}$

f. $\lim_{(x,y) \rightarrow (0,0)} \frac{x-2y}{x+y}$

c. $\lim_{(x,y) \rightarrow (0,0)} \frac{y^2 - x^2}{y^2 + x^2}$

g. $\lim_{(x,y) \rightarrow (0,0)} \frac{2y^4 x}{y^8 + 6x^2}$

d. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 y}{x^6 + y^2}$

3. In a robotic path planning the robot's position error is given by the function $f(x, y) = \frac{x^2 y - y^3 + \cos(xy)}{x^2 + y^2 + 1}$. Compute $\lim_{(x,y) \rightarrow (0,-1)} f(x, y)$.

4. Check whether the following functions are continuous or not

a. $f(x, y) = x^2 + y^2$ at $(1, 2)$

c. $f(x, y) = \begin{cases} \frac{xy}{x^2 + 5y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$

b. $f(x, y) = \begin{cases} \frac{1}{1+e^x} + y^2, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$

5. The electrical potential in a circuit is $V(x, y) = \frac{x^2 - y^2 + \sin(xy)}{x^2 + y^2 + 1}$. Check whether $V(x, y)$ is continuous at $(1, -1)$ or not.

6. If $z = x^3 y^2 + 3xy$, find the values of $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$, $\frac{\partial^2 z}{\partial x^2}$ and $\frac{\partial^2 z}{\partial y^2}$.

7. If $u = e^{xy+yz+zx}$, find the value of $\frac{\partial^3 u}{\partial x \partial y \partial z}$.

8. Compute $f_{xy}(0, 0)$ and $f_{yx}(0, 0)$ for the function $f(x, y) = \begin{cases} \frac{x^3 y}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$

9. Find $\frac{\partial x}{\partial u}$ and $\frac{\partial y}{\partial u}$, where the equations $u = x^2 - y^2$ and $v = x^2 - y$ define x and y as functions of the independent variables u and v , the partial derivatives exist. Also, if $s = x^2 + y^2$, find $\frac{\partial s}{\partial u}$.

10. Let $f(x, y, z) = x^3 y^2 + yz^4$ and $x = s^2$, $y = st^2$, $z = s^2 t$. Using the chain rule, compute $\frac{\partial f}{\partial s}$.

11. If $u = \sec^{-1}\left(\frac{x^3 - y^3}{x - y}\right)$, then find $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$.

12. Let $u = \frac{y^3 - x^3}{x^2 + y^2}$ prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = u$ and hence show that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 0$.

13. Use chain rule to find the derivative of $z = e^{xy}$ w.r.t t along the path $x = \cos t$, $y = \sin t$. Also find $\frac{dz}{dt}$ at $t = \frac{\pi}{2}$.

14. Suppose that a program's execution time depends on CPU cycles (x) and the memory usage (y) modeled by the equation $T(x, y) = x^2 + y^2$; where $x = p^2 - 1$ (CPU cycles depending on input parameter p) and $y = p^3 - p$ (memory usage depending on input parameter p). Use Chain rule of

partial differentiation to find the total derivative $\frac{dT}{dp}$; i.e., how execution time changes as we change the input parameter p .

15. Find the total differentiation of the following functions:

a. $z = \tan^{-1}\left(\frac{x}{y}\right)$; $(x, y) = (0, 0)$ b. $u = \left(xz + \frac{x}{z}\right)^y$; $z \neq 0$

16. Given transformation $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$ and $z = r \cos \theta$. Find the Jacobian.

17. In a computer graphics application, an image is transformed from original coordinates (x, y) to new coordinates (u, v) using the transformation: $u = 2x + y$ and $v = x - 2y$. Find the Jacobian.

Answers:

1.

2. a. 0 b. 0 e. 0

3. 1

4. a. Continuous b. Discontinuous c. Not continuous

5. Continuous

6. $\frac{\partial z}{\partial x} = 3x^2y^2 + 3y$, $\frac{\partial z}{\partial y} = 2x^3y + 3x$, $\frac{\partial^2 z}{\partial x^2} = 6xy^2$ and $\frac{\partial^2 z}{\partial y^2} = 2x^3$.

7. $e^{xy+yz+zx} [2(x+y+z) + (x+y)(x+z)(y+z)]$

8. $f_{xy}(0, 0) = 0$ and $f_{yx}(0, 0) = 0$

9. $\frac{\partial s}{\partial u} = \frac{1+2y}{1-2y}$

10. $\frac{\partial f}{\partial s} = 8s^7t^4 + 9s^8t^6$

11. $xu_x + yu_y = \frac{2(x^2+xy+y^2)}{|x^2+xy+y^2|\sqrt{(x^2+xy+y^2)^2-1}}$

13. $\frac{dz}{dt} = -1$

14. $\frac{dT}{dp} = 4p(p^2 - 1) + 2(p^3 - p)(3p^2 - 1)$

15. a. $dz = \frac{ydx - xdy}{x^2 + y^2}$ b. $du = \left(xz + \frac{x}{z}\right)^y \left[\ln\left(xz + \frac{x}{z}\right) dy + \frac{y}{xz + \frac{x}{z}} \left(\left(z + \frac{1}{z}\right) dx + x \left(1 - \frac{1}{z^2}\right) dz \right) \right]$

16. $r^2 \sin \theta$

17. -5